

Nuno de Sousa

Subject Matter Expert in Quantitative Modelling, Pricing & Forecasting | Data Scientist

Quantitative Strategy, Pricing, Forecasting & AI

PhD | Executive MBA

Based in Oporto, Portugal
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Portuguese | Male
Born 28 July 1982

nunodsousa.github.io/cv | LinkedIn | GitHub | ORCID

English CV

EXECUTIVE PROFILE

Data Science and AI leader with **over 15 years of experience** combining **quantitative expertise** with **business strategy and organizational leadership**. Currently works as a **Data Scientist and Subject Matter Expert**, developing **pricing and forecasting solutions** for complex, data-sparse markets. Has led **multidisciplinary teams of more than 10 professionals** and delivered **strategic analytics and AI initiatives** for international clients across energy, finance, chemicals and industrial sectors. Holds a **PhD in Theoretical Physics** and an **Executive MBA**, bringing together analytical rigour, executive communication and a growing focus on **strategy, management and business leadership**.

LEADERSHIP AT A GLANCE

Strategy & Delivery
Data and AI strategy, solution architecture, prioritisation, business-technology alignment and delivery of analytics initiatives.

People Leadership
Leadership of multidisciplinary teams of more than 10 professionals, cross-functional coordination and technical direction.

Stakeholders & Growth
Executive communication, strategic advisory, pre-sales and client-facing solution design.

PROFESSIONAL EXPERIENCE

Subject Matter Expert in Quantitative Modelling, Pricing & Forecasting | Data Scientist — Mar 2026–Present
AuctionConnect — Oporto / Remote

- Defines the analytical approach for nowcasting and forecasting marine bunker fuel prices across global ports, aligning model development with commercial priorities and market needs.
- Transforms fragmented market data, economic indicators and pricing signals into actionable intelligence for commercial planning, risk assessment and strategic decision-making.
- Develops robust statistical and machine learning frameworks for complex, data-sparse markets, balancing analytical accuracy with business applicability.
- Communicates model assumptions, outputs and market implications to business stakeholders and contributes specialist knowledge to the long-term evolution of pricing products.

Data & AI Lead | Subject Matter Expert — Nov 2019–Mar 2026
DataJuicers and Simia-Tech — Madrid / Oporto / Remote

- Led cross-functional teams of more than 10 data scientists, engineers and analysts, aligning business objectives, analytical rigour and timely delivery for international clients.
- Architected production-grade Data Science and AI platforms and advanced models for pricing optimisation, demand forecasting, inventory planning and market intelligence.
- Served as the business-technology interface and strategic advisor to senior stakeholders, translating analytical outputs into clear decision support.
- Supported pre-sales through solution design, technical validation and client-facing presentations, helping secure and expand long-term engagements.

Quantitative Analyst — Apr 2019–Oct 2019
Arfima Trading — Madrid

Designed and deployed automated futures trading strategies from end-of-session to high-frequency horizons. Built statistical and machine learning models, a custom backtesting framework and integrated risk management protocols.

Data Scientist — Jun 2017–Mar 2019
DataJuicers — Madrid

Delivered big-data, time-series forecasting and machine learning solutions across luxury, retail, services and industry. Led a TensorFlow computer-vision recommendation project recognised by Google and contributed quantitative analysis to hedge-fund projects.

Researcher / Lecturer — 2009–Jun 2017
Universidad Autónoma de Madrid & Donostia International Physics Center — Spain

Taught Physics and Informatics Engineering, supervised research students, and conducted research in Computational Physics, Electrodynamics and Mathematical Methods.

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EDUCATION

Executive MBA Oct 2024–Oct 2026
Católica Porto Business School - CPBS

- Strategic leadership and decision-making: executive decisions, competitive strategy and business model innovation.
- Corporate finance and risk management: financial analysis, valuation, capital allocation and enterprise risk.
- Operations and organisational performance: operational excellence, performance management and large-scale transformation.
- Executive communication and governance: negotiation, stakeholder management and corporate governance frameworks.

PhD in Theoretical Physics Sep 2010–Oct 2014
Universidad Autónoma de Madrid
Condensed Matter Physics and Nanotechnology. Dissertation: *Light scattering in disordered and nonreciprocal media*. Awarded *Sobresaliente Cum Laude*, the maximum classification.

Master in Photonics Sep 2009–Jun 2010
Universidad Autónoma de Madrid
GPA 8.8/10. Thesis on light-emission statistics in correlated random photonic nanostructures.

Licentiate Degree in Physics
Universidade do Porto

LEADERSHIP, BUSINESS & TECHNICAL CAPABILITIES

Leadership & Management

Strategy & Delivery
Data and AI strategy, solution architecture, initiative prioritisation, delivery governance and business-technology alignment.

People Leadership
Multidisciplinary team leadership, cross-functional coordination, technical direction, mentoring and quality ownership.

Stakeholders & Growth
Executive stakeholder communication, strategic advisory, pre-sales, negotiation and client-facing solution design.

Business Applications
Pricing optimisation, demand forecasting, inventory planning, market intelligence and quantitative trading.

Quantitative Modelling
Pricing models, time-series forecasting (ARIMA, ETS, state space and Prophet), Bayesian modelling, simulation, optimisation, survival analysis and customer lifetime value.

Machine Learning & AI
scikit-learn, XGBoost, TensorFlow, LLM applications, RAG, model evaluation and agentic workflows.

Data & Production Systems
Python, SQL, pandas, NumPy, statsmodels, Docker, Git, Azure and workflow orchestration with n8n.

RESEARCH, TEACHING & RECOGNITION

22	~50	5	8
peer-reviewed publications	international oral presentations	theses supervised	science research projects

- Co-director of the *AI for Marketing* programme at Católica Porto Business School.
- Led the 2017 Dress Recommendation System, recognised by Google for its application of TensorFlow and computer vision.
- Supervised three master's theses and two bachelor's dissertations across Physics and Informatics Engineering.
- Latest publication: A. P. V. Oliveira, S. C. Silva and N. de Sousa, *Simplicity over sophistication? Rule-based chatbots lead to higher return rates than LLMs in retail*. *The International Review of Retail, Distribution and Consumer Research* (2026).

PROFESSIONAL DEVELOPMENT & LANGUAGES

Selected certifications

Python for Financial Analysis and Algorithmic Trading - Udemy, 2018
Machine Learning Specialization - University of Washington / Coursera, 2016–2017
Applied Plotting, Charting and Data Representation in Python - University of Michigan / Coursera, 2017
Inferential Statistics - Duke University / Coursera, 2017

Languages

Portuguese - Native
English - Certified C1+ (Aug 2024)
Spanish - 12 years living in Spain

Full professional and academic record: nunodsousa.github.io/cv/